

DECODER

AIM:

To construct and verify 3x8 decoder using the IC 74136.

APPARATUS REQUIRED:

- (i) 3x8 decoder - IC 74138
- (ii) Connecting wires

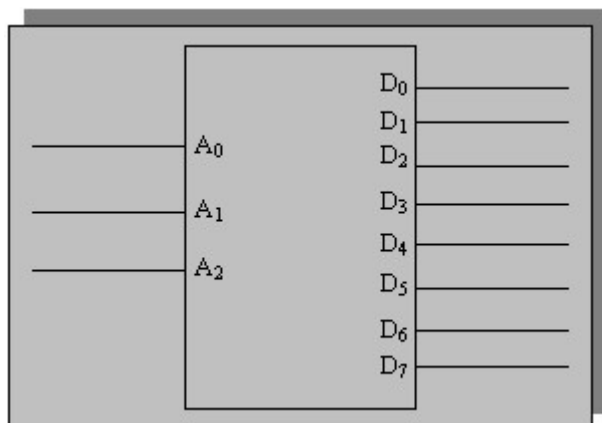
THEORY:

Discrete quantities of information are represented in digital systems by binary codes. A binary code of n bit can represent n th power of 2 information. A decoder is a combinational circuit that converts binary information from n input lines. These decoders are called n to m line decoders. Where m may be lesser than or equal to n th power of 2. Their purpose is to generate n th power of 2 minterms of n input variables. The name decoder is also used in conjunction with other code convertors; an eg. is BCD to seventh segment decoder. The 3 inputs are decoded into 8 outputs, each representing one of the minterms of the 3 input variables. A particular application of this decoder is octal to binary conversion. The input variables represent a binary number and the outputs represent 8 digit number in octal number system.

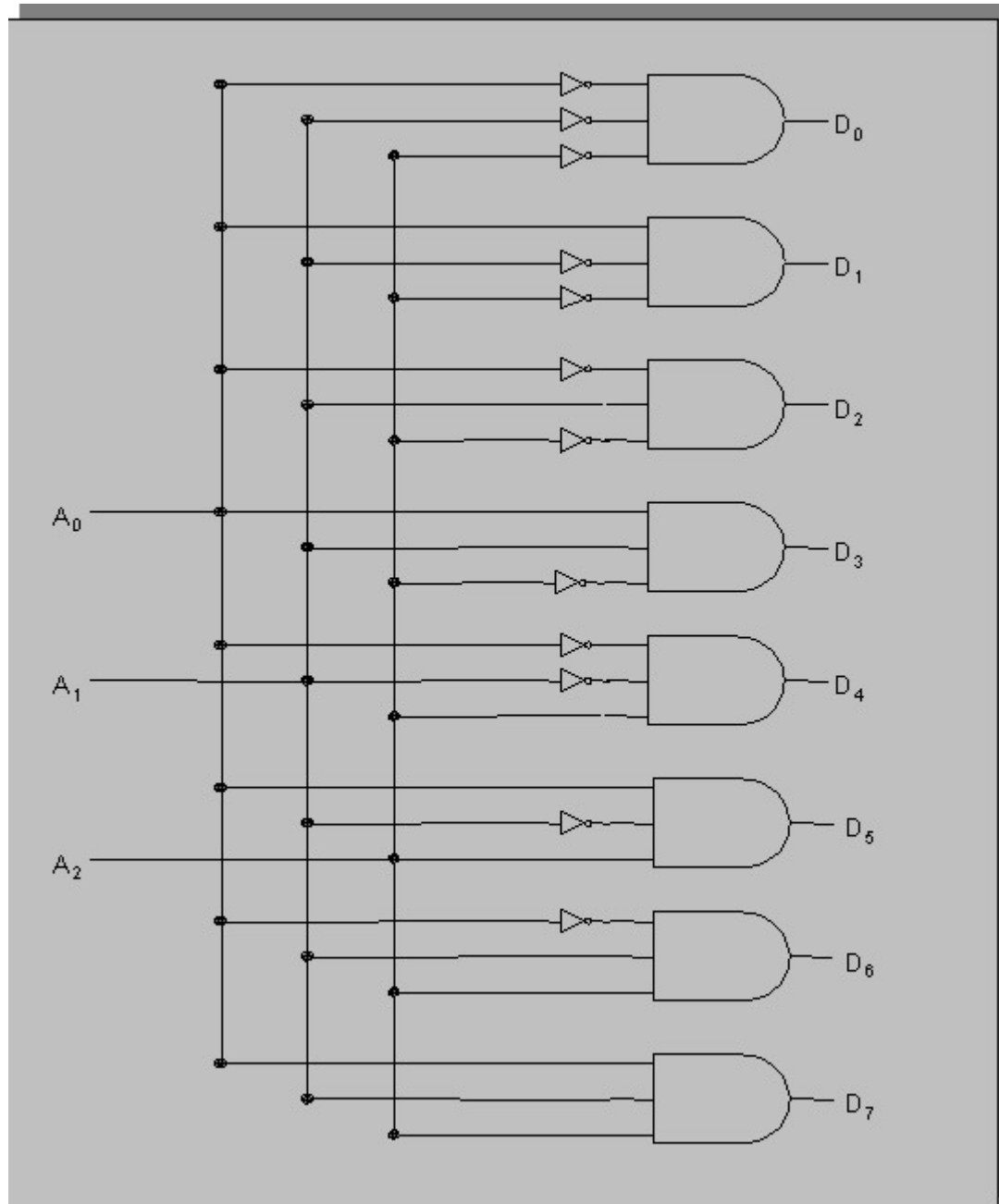
PROCEDURE:

The 3x8 decoder is mainly used for binary to octal conversion. The circuit connections are given as shown in circuit diagram. Inputs are given to the 3 invertors. Then each of the AND gates generates one of the minterms for the inputs. Then the decimal decoded output is verified.

PIN DIAGRAM:(IC 74138)



CIRCUIT DIAGRAM:



FUNCTION TABLE:

INPUTS					OUTPUTS							
ENABLE		SELECT										
G1	G2	A2	A1	A0	D0	D1	D2	D3	D4	D5	D7	D8
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	H	L	L	L	L	H	H	H	H	H	H	H
H	H	L	L	H	H	L	H	H	H	H	H	H
H	H	L	H	L	H	H	L	H	H	H	H	H
H	H	H	H	H	H	H	H	L	H	H	H	H
H	H	H	L	L	H	H	H	H	L	H	H	H
H	H	H	L	H	H	H	H	H	H	L	H	H
H	H	H	H	L	H	H	H	H	H	H	L	H
H	H	H	H	H	H	H	H	H	H	H	H	L

Result:

Thus the 3x8 decoder is constructed and verified using the digital IC 74138.

OCTAL TO BINARY ENCODER

AIM:

To realise and verify the truth table of octal to binary encoder using IC74148 digital IC.

APPARATUS REQUIRED:

S.no	Name of the equipment	Specification	No.of quantity
1.	8X3 encoder	IC 74148	1
2.	Trainer kit	-	1
3.	Connecting wires	-	As required

PIN CONFIGURATION :

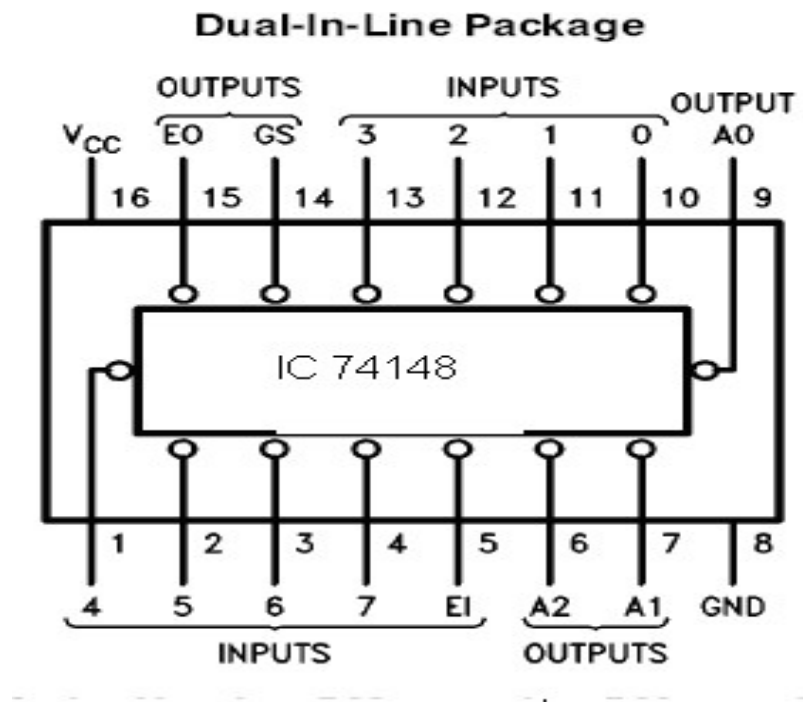


Fig.1

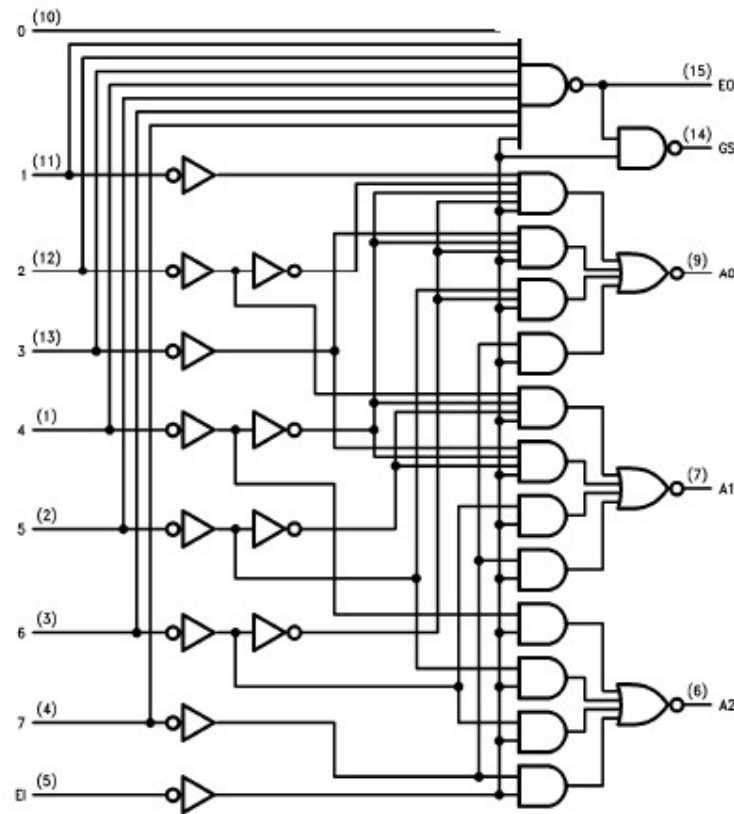
Function table

DM54148

Inputs									Outputs				
E1	0	1	2	3	4	5	6	7	A2	A1	A0	GS	E0
H	X	X	X	X	X	X	X	X	H	H	H	H	H
L	H	H	H	H	H	H	H	H	H	H	H	H	L
L	X	X	X	X	X	X	X	L	L	L	L	L	H
L	X	X	X	X	X	X	L	H	L	L	H	L	H
L	X	X	X	X	X	L	H	H	L	H	L	L	H
L	X	X	X	L	H	H	H	H	H	L	L	L	H
L	X	X	L	H	H	H	H	H	H	L	H	L	H
L	X	L	H	H	H	H	H	H	H	H	L	L	H
L	L	H	H	H	H	H	H	H	H	H	H	L	H

H = High Logic Level, L = Low Logic Level, X = Don't Care

Logic Diagram



THEORY

An encoder is a digital function that produces reverse operation from that of a decoder. An encoder has 2^n input lines and n corresponding binary numbers. It is constructed with OR gates.

The encoder assume that only one input lines can be equal to 1 at any time ,otherwise the circuit has no meaning.The circuit has 8 inputs and can have $2^8=256$ possible input combination. Only 8 of these combination are in "dont care" condition. This type of encoder available in IC form is called priority encoder. The circuit has active low inputs and active low outputs. The enable input and carry outputs which are also active low are used to cascade circuits which are used to handle more inputs.

PROCEDURE:

The connections are given as shown in the pin diagram. The inputs are given to the corresponding pin as in the truth table and the outputs are verified.

RESULT:

Thus the truth table of octal to binary encoder is realised and verified using IC 74148 .